

In the same file, remove the pre-existing custom rule set by selecting it and pressing **Ctrl+K**.

To understand pre-existing rules, open the rule file *exploit.rules* under */etc/snort/rules*. For example, consider the following rule:

```
alert tcp $EXTERNAL_NET any ->
$HOME_NET 22 (msg:"EXPLOIT ssh
CRC32 overflow /bin/sh"; flow:to_
server,established; content:"/
bin/sh"; reference:bugtraq,2347;
reference:cve,2001-0144;
reference:cve,2001-0572;
classtype:shellcode-detect; sid:1324;
rev:6;)
```

This Snort rule detects an SSH exploit by matching the string `/bin/sh` in TCP traffic targeting port 22. It identifies the attack as a CRC32 overflow and references Bugtraq ID 2347 and CVEs 2001-0144, 2001-0572, classifying it as a possible shellcode attack. Bugtraq is a mailing list/database for reporting and discussing security vulnerabilities, while a CVE (Common Vulnerabilities and Exposures) is a standardised identifier for publicly disclosed security flaws. These CVEs from 2001 describe a buffer overflow in the SSH daemon's CRC32 compensation attack detector, where crafted packets could trigger memory corruption, allowing remote attackers to execute arbitrary code and spawn a shell on the target system.

Now, validate the *snort.conf* file with the following command (replace *enp0s3* in the following command with the correct interface found using the command from your system `$ ip link show`).

```
$ sudo snort -T -i enp0s3 -c /etc/
snort/snort.conf
```

Editing *snort.conf*

The *snort.conf* file is the main configuration file of Snort. Any change

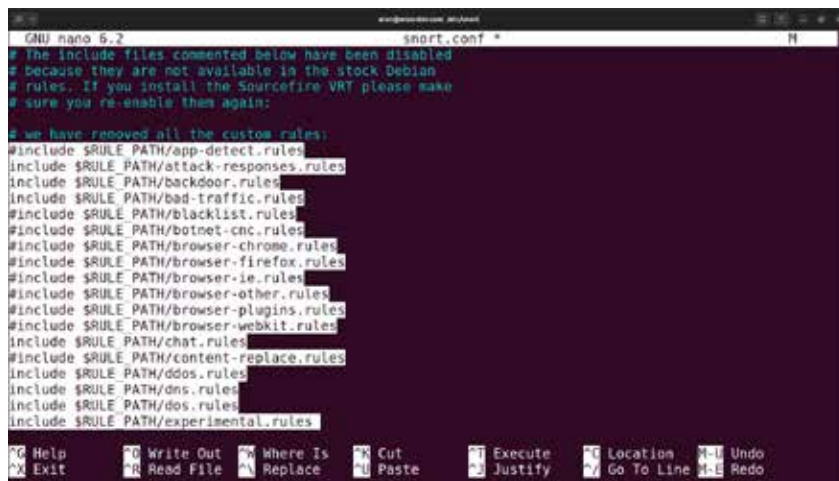


Figure 3: Removing existing custom rules

in settings, network variables, or rule paths is managed here. It is important to open this file only from its original directory (*/etc/snort*). This ensures that the correct file is being edited. After every change made in this file, the configuration should be validated. Validation helps confirm that there are no syntax errors or missing values before Snort is started.

The validation command is:

```
$ sudo snort -T -i enp0s3 -c /etc/snort/
snort.conf
```

Writing custom rules in *local.rules*

Snort rules follow a fixed format. A simple rule looks like this:

```
action protocol source_ip source_port
-> destination_ip destination_port
(options)
```

- **action:** What Snort should do when the rule matches (example: alert).
- **protocol:** The protocol to check, such as TCP, UDP, or ICMP.
- **source_ip and source_port:** The address and port of the traffic origin (any can be used for all).
- **destination_ip and destination_port:** The address and port of the target system.

- **options:** Extra instructions in brackets, such as the message (msg), rule ID (sid), and version (rev).

After removing the custom rules that are available in *snort.conf*, move to the *rules* directory and open the *local.rules* file using:

```
$ cd rules
$ sudo nano local.rules
```

The rules demonstrate detection of FTP traffic using both static and dynamic approaches.

Static rule

```
alert tcp any any -> any 21 (msg:"FTP
Traffic Detected"; sid:1000001; rev:1;)
```

This rule is always active and triggers whenever TCP traffic is observed on port 21, which is the default FTP control port. Since it is static, it will continuously monitor and alert FTP connections from the start of the FTP session until the session exits.

Dynamic rules with flowbits

- *Detecting the FTP USER command*

```
alert tcp any any -> any 21 (msg:"FTP
USER command seen"; content:"USER ";
nocase; flowbits:set,ftp_user_seen;
sid:500001;)
```